Assessment–1: Assignment and Viva Voce (10 Marks)

This assessment consists of a **hands-on assignment** followed by an **iviva voce** to evaluate conceptual clarity and practical understanding.

Note:

- The **maximum group size is two**.
- **Evaluation will be individual**, irrespective of group work.

A. Evaluation Rubrics

The assignment and viva will be evaluated based on the following criteria:

1. **Understanding of the Dataset and Data Sources**
    - Nature of the data
    - Type of attributes
    - Real-world relevance
2. **Input and Output Specification**
    - Clearly defined features and target variable
    - Problem formulation
3. **Mathematical Modeling**
    - Mathematical formulation of the problem
    - Variables involved
    - Optimization techniques used
4. **Computational Steps**
    - Step-by-step explanation of the selected model
    - Pipeline clarity
5. **Coding Proficiency**
    - Code readability
    - Proper use of libraries and functions

B. Problem Statement

A **healthcare analytics company** has collected a dataset to predict whether a patient is at **high risk of developing a chronic disease**.

Dataset Description

The dataset contains the following attributes:

- `Patient_ID`
- `Age`
- `Gender`
- `Blood_Pressure` *(contains missing values)*
- `Cholesterol_Level` *(contains extreme values / outliers)*
- `Hospital_Visits_Per_Year`
- `Smoking_Status`

- `Exercise_Level`
- `Medical_Notes` *(free-text data)*
- `Wearable_Sensor_Data` *(time-series data)*
- `Disease_Risk` *(binary target variable; highly imbalanced)*

C. Tasks to be Performed

Students are required to perform the following steps **systematically**:

1. **Data Understanding and Pre-processing**
   - Identify data types (numerical, categorical, text, time-series).
   - Perform **encoding** wherever required (with justification).
2. **Missing Value Handling**
   - Apply an appropriate missing value imputation technique.
   - **Justify the choice of method** based on data characteristics.
3. **Class Imbalance Handling**
   - Address the class imbalance problem using a suitable technique.
   - **Justify why the chosen method is appropriate**.
4. **Outlier Detection and Treatment**
   - Detect outliers in relevant features.
   - Apply a suitable outlier handling method with justification.
5. **Feature Scaling**
   - Perform **normalization or standardization**.
6. **Feature Selection / Dimensionality Reduction**
   Apply and compare the following methods:
   - Variance-based feature selection
   - Information Gain
   - Forward Feature Selection
   - Principal Component Analysis (PCA)
7. **Model Building**
   - Apply **Logistic Regression** for classification..
8. **Comparative Study**
   - Compare **Logistic Regression performance** with:
     - No feature selection
     - Variance method
     - Information Gain
     - Forward Selection
     - PCA
   - Present results in a **tabular format** using **F1-Score** as the primary evaluation metric.

D. Expected Deliverables

1. **Well-commented Source Code**
2. **Viva Voce**